Qn 5	Answer	Marks
(a)(i)	The potential difference between X and Y, $\Delta V = E\Delta x = (4000)(0.040)$ $= 160\ V$ (Allow: -160 V). Potential at X is higher.	**[1]** **[1]** **[1]**
(a)(ii)	(By conservation of energy, the work done by an external force to move the charge from X to Y = increase in electrical potential energy of the system.) The work done by an external force to move the charge from X to Y $W = \Delta V \times q = (-160)(1.20 \times 10^{-9}) = -1.92 \times 10^{-7}$ J [Remark: if the student gives positive answer – allow 1 of the 2 marks]	**[1]** **[1]**
(b)(i)	Number of electrons per second $N = \frac{I}{q} = \frac{7.20 \times 10^{-3}}{1.6 \times 10^{-19}}$ $= 4.50 \times 10^{16}$	**[1]** **[1]**
(b)(ii)	Gain in K.E. of the electrons = Loss in EPE ½ mv² 0 = e⊗V $v = \sqrt{\left(\frac{2}{m}\right)(\Delta V)(e)}$ $= \sqrt{\left(\frac{2}{9.11 \times 10^{-31}}\right)(12 \times 10^{3})(1.6 \times 10^{-19})}$ $= 6.49 \times 10^{7}\ m s^{-1}$	**[1]** **[1]**
	Max Marks	**9**

Qn 6	Answer	Marks